

A Unified Predefined-Time Convergent and Robust ZNN Model for Constrained Quadratic Programming

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I. INTRODUCTION

Abstract—A variety of realistic industrial problems can be constructed into quadratic programming (QP) problems, especially their time-varying versions. A zeroing neural network (ZNN) as a good approach for dynamic problems can solve QP problems subject to equality constraints in the past. In this article, we propose a unified predefined-time convergent and robust ZNN (PTCR-ZNN) model for solving time-varying QP problems subject to equality or inequality constraints. Compared with the normal ZNN model, the PTCR-ZNN model mainly has advantages in the following three aspects: 1) solving QP problems with or without inequality constraints in a unified model; 2) converging to the optimal solution of QP problems within a predefined time that can be determined in advance; and 3) resisting many external noises with tiny and predictable residual error. These improvements have been rigorously proved in theory. By conducting both qualitative and quantitative simulations with comparisons, the superior properties of the PTCR-ZNN model are further validated. Finally, the application of the PTCR-ZNN model to image fusion task illustrates the efficiency together with its applicability.

Index Terms—Image fusion, predefined-time convergence (PTCR), quadratic programming (QP), robustness, zeroing neural network (ZNN).

QUADRATIC programming (QP) plays an important part in various applications, including image processing [1], [2], and robot control [3], [4]. As a core analytical tool in many realistic problems, the QP problem is worth being investigated both for mathematical and practical purposes. Besides, solving time-varying QP (TVQP) problems has attracted more researchers' attention recently for its difficulty and broad usage.

Numerical approaches, including iterative algorithms [5], [6], are well-accepted ways for solving QP problems. However, the minimum time complexity of these approaches is no less than the third power of Hessian matrix's size, which have significantly limited numerical methods on solving large-scale QP problems in real time. When using numerical methods to solve TVQP problems, such time-consuming operations need to be finished in every sampling gap, and the higher accuracy we require, the shorter gap we have. Therefore, TVQP problems are even difficult for numerical ways.

Recently, neural dynamic approaches have been broadly investigated to tackle QP problems, thanks to their parallel structure, as a typical example of which the recurrent neural networks (RNNs) can be speeded up by using large-scale digital circuit or parallel computing for their notable parallel properties. A gradient-based neural network (GNN) as a type of RNN is early used in this research field [7]–[9]. However, the GNN can only perform well on static QP problems and always has nonvanished residual error on solving TVQP problems because of using the scalar-type energy function. Since the GNN is unable to obtain the time-dependent problems' precise solutions, later on, a zeroing neural network (ZNN) as a novel neural dynamic was presented in [10]–[12]. Compared with the GNN, the ZNN takes advantage of time-derivative information of input coefficients and, thus, can converge to precise solutions of TVQP problems [13] without large lagging errors.

For ZNN models, one important thing that researchers focus on is the convergence time of the computation error converging to zero. ZNN models using classic activation functions (AFs) in [14] often need infinite convergence time. The authors of [15] and [16] use special AFs, including the well-known sign-bi-power (SBP) function in ZNN models, making them finite-time convergent. Nevertheless, the convergence time of these

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finite-time convergent models is restricted by their initial errors, which are less controllable and reliable than predefined-time convergent models with their convergence time determined in advance [17], [18]. Another element we should study about ZNN models is their robustness, since various noises in real application environment are inevitable [19]. In [20] and [21], different integration-enhanced ZNN models are proposed, and they possess superior noise suppression ability. Although, in real world, the performance of a comprehensive system such as ZNN is often determined by its weakest spot, there are few works on ZNN models for QP problems subject to equality or inequality constraints that provide a model with both fast convergence rate and powerful robustness. For example, in [10], TVQP problems are only considered with equality constraints (TVQPE) and not with inequality constraints. More general TVQP problems with equality and inequality constraints (TVQPEI) are solved by a ZNN model in [13] but with slower exponential convergence rate. Recently, ZNN models that can tackle general TVQPEI problems and have accelerated finite-time or predefined-time convergence speed have been proposed in [18] and [22]. However, robustness performance is ignored in these two works again. Therefore, following the inspiration of designing a ZNN model being predefined-time convergent as well as noise resistant, we propose a unified predefined-time convergent and robust zeroing neural network (PTCR-ZNN) model in this article for solving TVQP problems subject to equality or inequality constraints.

In addition, we have selected two real-world applications to validate efficacy of our PTCR-ZNN mode. One application is to achieve image denoising by image fusion, and another application is to realize inverse kinematics control of a redundant robotic manipulator. In [2], the image fusion task is modeled into a QP problem and is solved by a GNN model, and one disadvantage of this GNN method is its low convergence speed when compared to predefined-time ZNN models. For the robot manipulator control task, there are various schemes such as obtaining Jacobian matrix pseudoinversion [23], solving the TVQPE problem [4], and so on. However, most of these redundant robot manipulator control methods cannot handle joint limits, which is necessary in practice for avoiding obstacles or physical limits. In this article, our PTCR-ZNN model will overcome these shortcomings and demonstrate its superiorities in these applications.

The structure of this article is outlined as follows. The formulation of PTCR-ZNN model's targeting problems and some basic knowledge about solving these optimizations problems are shown in Section II. Section III presents dynamic formulas of the PTCR-ZNN model and our newly proposed activation function (piecewise predefined-time activation function—PPAF). Sections IV and V rigorously analyzed the convergence and robustness performance of the PTCR-ZNN model, respectively. Section VI provides numerical simulations on the PTCR-ZNN model solving QP problems, which have verified conclusions in Sections IV and V. In Section VII, the PTCR-ZNN model is applied to image fusion with success and excellent effect. In Section VIII, we successfully apply the PTCR-ZNN model to control a redundant robot arm with

joint limits considered. Finally, Section IX concludes this article.

The major contributions of this article are pointed as follows.

- 1) A unified PTCR-ZNN model is proposed to solve TVQP problems with or without inequality constraints.
- 2) The PTCR-ZNN model is predefined-time convergent in noise free environment and has superior noise-tolerant ability in noisy environment.
- 3) Theoretical analyses are provided to support the convergence and robustness performance of the PTCR-ZNN model especially with dynamic unknown additive noises.
- 4) Numerical experiments validate the efficacy of the proposed PTCR-ZNN model for TVQP problems, where our theoretical conclusions about the PTCR-ZNN model are quantitatively verified.

II. PRELIMINARIES

A. Problem Formulation

In this article, we are mainly concerned with TVQP problems subject to equality or inequality constraints. Specifically, there are two types of TVQP problems: one is TVQP problems with equality constraints (TVQPE), and the other is TVQP problems with equality and inequality constraints (TVQPEI). In this article, we assume that each TVQP has a unique optimal solution. First, let us consider the standard form of the TVQPEI problem as

$$\min \mathbf{x}(t)^T \mathbf{S}(t) \mathbf{x}(t) / 2 + \mathbf{p}^T(t) \mathbf{x}(t) \quad (1)$$

$$\text{s.t. } \mathbf{A}(t) \mathbf{x}(t) = \mathbf{b}(t) \quad (2)$$

$$\mathbf{C}(t) \mathbf{x}(t) \leq \mathbf{d}(t) \quad (3)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ is an unknown vector and should be obtained, $\mathbf{S}(t) \in \mathbb{R}^{n \times n}$ represents a Hessian matrix that is symmetric and positive definite, $\mathbf{A}(t) \in \mathbb{R}^{m \times n}$ is a full rank matrix, and $\mathbf{C}(t) \in \mathbb{R}^{q \times n}$. Other coefficient vectors are $\mathbf{p}(t) \in \mathbb{R}^n$, $\mathbf{b}(t) \in \mathbb{R}^m$, and $\mathbf{d}(t) \in \mathbb{R}^q$. Obviously, with these coefficient matrices and vectors all being time varying, the optimal solution $\mathbf{x}^*(t) \in \mathbb{R}^n$ to the TVQPEI problems is also changing with time. Then, we consider the standard form of the TVQPE problem, which can be expressed as follows:

$$\begin{aligned} \min \mathbf{x}(t)^T \mathbf{S}(t) \mathbf{x}(t) / 2 + \mathbf{p}^T(t) \mathbf{x}(t) \\ \text{s.t. } \mathbf{A}(t) \mathbf{x}(t) = \mathbf{b}(t) \end{aligned} \quad (4)$$

where $\mathbf{S}(t)$, $\mathbf{p}(t)$, $\mathbf{A}(t)$, and $\mathbf{b}(t)$ share the same definitions with TVQPEI (1), (2).

Apparently, the major difference between TVQPEI and TVQPE lies in that TVQPE has no inequality constraints. In order to lay the foundation for the rest of this article, we can transform TVQPE into the TVQPEI problem by adopting $\mathbf{C}(t) = [\mathbf{I}_n, -\mathbf{I}_n]^T \in \mathbb{R}^{2n \times n}$ and $\mathbf{d}(t) = [\infty_n^T, \infty_n^T]^T$, where $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ is an identity matrix and ∞_n is an n -dimensional column vector, with each element being positive infinity. Because positive infinity is hard to represent in computer and digital circuit, a large enough positive real number is usually adopted to denote it.

B. Optimization Theory

According to Section II-A, we conclude that the TVQPEI problem is the most generalized form among TVQP problems, since TVQPEI problems can be easily reduced to the special case of TVQPEI problems. Therefore, we apply the Karush–Kuhn–Tucker (KKT) theory to TVQPEI (1)–(3) and obtain the following lemma.

Lemma 1: Vector $\mathbf{x}^*(t) \in \mathbb{R}^n$ becomes the optimal solution of the TVQPEI problem (1)–(3) if and only if the Lagrangian multipliers $\boldsymbol{\rho}^*(t) \in \mathbb{R}^m$ and $\boldsymbol{\lambda}^*(t) \in \mathbb{R}^q$ exist such that they satisfy the following KKT conditions [1], [18], [24], [25]:

$$\begin{cases} \mathbf{S}(t)\mathbf{x}^*(t) + \mathbf{p}(t) + \mathbf{A}^T(t)\boldsymbol{\rho}^*(t) + \mathbf{C}^T(t)\boldsymbol{\lambda}^*(t) = \mathbf{0} \\ \mathbf{A}(t)\mathbf{x}^*(t) - \mathbf{b}(t) = \mathbf{0} \\ \boldsymbol{\lambda}^*(t) \geq \mathbf{0}, \mathbf{d}(t) - \mathbf{C}(t)\mathbf{x}^*(t) \geq \mathbf{0} \\ \boldsymbol{\lambda}^{*T}(\mathbf{d}(t) - \mathbf{C}(t)\mathbf{x}^*(t)) = \mathbf{0} \end{cases} \quad (5)$$

where vectors $\boldsymbol{\rho}^*(t)$ and $\boldsymbol{\lambda}^*(t)$ are unknown and should be obtained to meet constraints (2) and (3). Therefore, $\mathbf{y}^*(t) = [\mathbf{x}^*(t)^T, \boldsymbol{\rho}^*(t)^T, \boldsymbol{\lambda}^*(t)^T]^T \in \mathbb{R}^{n+m+q}$ is the solution vector of the TVQPEI problem.

In general, ZNN models are suitable for equality-constrained instead of inequality-constrained QP problems; thus, the vector-type perturbed Fischer–Burmeister (PFB) function in [1], [18], and [25] is introduced for converting the inequality constraints in (5). The PFB function is defined as follows:

$$\psi_{\text{PFB}}(\mathbf{a}, \mathbf{c}) = \mathbf{a} + \mathbf{c} - \sqrt{\mathbf{a} \circ \mathbf{a} + \mathbf{c} \circ \mathbf{c} + \boldsymbol{\delta}^+}, \quad \boldsymbol{\delta}^+ \rightarrow \mathbf{0}_+ \quad (6)$$

where $\mathbf{a}$, $\mathbf{c}$, and $\boldsymbol{\delta}^+$ are all column vectors with the same dimension, and $\boldsymbol{\delta}^+$ is called the perturbed item. In addition, symbol $\circ$ denotes the Hadamard product. According to [25], we have the following lemma.

Lemma 2: Considering the scalar type of PFB function (6), we can conclude that

$$\psi_{\text{PFB}}(a, c) = 0 \Leftrightarrow a \geq 0, c \geq 0, ac = \boldsymbol{\delta}^+ / 2$$

where $a, c, \boldsymbol{\delta}^+ \in \mathbb{R}, \boldsymbol{\delta}^+ > 0$. Besides, $\psi_{\text{PFB}}(a, c)$ is smooth with regard to a and c .

Therefore, the vector-valued PFB function (6) satisfies Lemma 2 by element. According to (5), we further have the following result.

Lemma 3: The optimal solution $\mathbf{y}^*(t) = [\mathbf{x}^*(t)^T, \boldsymbol{\rho}^*(t)^T, \boldsymbol{\lambda}^*(t)^T]^T \in \mathbb{R}^{n+m+p}$ satisfies the following equations:

$$\begin{cases} \mathbf{S}(t)\mathbf{x}^*(t) + \mathbf{p}(t) + \mathbf{A}^T(t)\boldsymbol{\rho}^*(t) + \mathbf{C}^T(t)\boldsymbol{\lambda}^*(t) = \mathbf{0} \\ \mathbf{A}(t)\mathbf{x}^*(t) - \mathbf{b}(t) = \mathbf{0} \\ \psi_{\text{PFB}}(\mathbf{d}(t) - \mathbf{C}(t)\mathbf{x}^*(t), \boldsymbol{\lambda}^*(t)) = \mathbf{0} \end{cases}$$

for every $\boldsymbol{\delta}^+ \rightarrow \mathbf{0}_+$ if and only if $\mathbf{x}^*(t)$ is the KKT point of TVQPEI (1)–(3).

The above equations can be reformulated as the following nonlinear equation:

$$\mathbf{f}(\mathbf{y}(t), t) = \mathbf{G}(t)\mathbf{y}(t) + \mathbf{u}(t) = \mathbf{0} \quad (7)$$

with

$$\mathbf{G}(t) = \begin{bmatrix} \mathbf{S}(t) & \mathbf{A}^T(t) & \mathbf{C}^T(t) \\ \mathbf{A}(t) & \mathbf{0} & \mathbf{0} \\ -\mathbf{C}(t) & \mathbf{0} & \mathbf{I} \end{bmatrix}, \mathbf{y}(t) = \begin{bmatrix} \mathbf{x}(t) \\ \boldsymbol{\rho}(t) \\ \boldsymbol{\lambda}(t) \end{bmatrix}$$

$$\mathbf{u}(t) = \begin{bmatrix} \mathbf{u}_1(t) \\ \mathbf{u}_2(t) \\ \mathbf{u}_3(t) \end{bmatrix} = \begin{bmatrix} \mathbf{p}(t) \\ -\mathbf{b}(t) \\ \mathbf{d}(t) - \sqrt{\mathbf{h}(t) \circ \mathbf{h}(t) + \boldsymbol{\lambda}(t) \circ \boldsymbol{\lambda}(t) + \boldsymbol{\delta}^+} \end{bmatrix}$$

where $\mathbf{h}(t) := \mathbf{d}(t) - \mathbf{C}(t)\mathbf{x}(t)$. Then, solving the TVQPEI problem is equivalent to finding a time-varying vector $\mathbf{y}(t)$ that satisfies (7). In the following content of this article, we will propose a PTCR-ZNN model to tackle such a problem in predefined time and with strong robustness.

III. PTCR-ZNN MODEL DESIGN

On the basis of the standard ZNN design routine, an error vector is defined as $\boldsymbol{\varepsilon}(t) = \mathbf{f}(\mathbf{y}(t), t)$, which is often defined to track the evolution of neural networks. Instead of using the conventional ZNN formula such as

$$\dot{\boldsymbol{\varepsilon}} = -\gamma \Psi(\boldsymbol{\varepsilon}(t)) \quad (8)$$

we adopt the following novel design formula [21], [26]:

$$\dot{\boldsymbol{\varepsilon}}(t) = -\gamma_1 \Psi_1(\boldsymbol{\varepsilon}(t)) - \gamma_2 \Psi_2 \left(\boldsymbol{\varepsilon}(t) + \gamma_1 \int_0^t \Psi_1(\boldsymbol{\varepsilon}(\tau)) d\tau \right) \quad (9)$$

where $\gamma_1 > 0$ and $\gamma_2 > 0$ are design parameters used to tune the convergence speed, and $\Psi_1(\cdot)$ and $\Psi_2(\cdot)$ are odd monotonically increasing AF arrays, which can be selected arbitrarily. By using design dynamic (9), $\mathbf{f}(\mathbf{y}(t), t)$ will be forced to converge to zero vector, and the optimal solutions are obtained at the same time. Expanding (9), we obtain the PTCR-ZNN model:

$$\dot{\mathbf{G}}(t)\mathbf{y}(t) + \mathbf{G}(t)\dot{\mathbf{y}}(t) + \dot{\mathbf{u}}(t) = -\gamma_1 \Psi_1(\mathbf{G}(t)\mathbf{y}(t) + \mathbf{u}(t)) - \gamma_2 \Psi_2 \left(\mathbf{G}(t)\mathbf{y}(t) + \mathbf{u}(t) + \gamma_1 \int_0^t \Psi_1(\mathbf{G}(\tau)\mathbf{y}(\tau) + \mathbf{u}(\tau)) d\tau \right)$$

which can be further reformulated as the following expression:

$$\mathbf{J}(t)\dot{\mathbf{y}}(t) = -\mathbf{M}(t)\mathbf{y}(t) - \mathbf{v}(t) - \gamma_1 \Psi_1(\mathbf{G}(t)\mathbf{y}(t) + \mathbf{u}(t)) - \gamma_2 \Psi_2 \left(\mathbf{G}(t)\mathbf{y}(t) + \mathbf{u}(t) + \gamma_1 \int_0^t \Psi_1(\mathbf{G}(\tau)\mathbf{y}(\tau) + \mathbf{u}(\tau)) d\tau \right) \quad (10)$$

where

$$\mathbf{J}(t) = \begin{bmatrix} \mathbf{S}(t) & \mathbf{A}^T(t) & \mathbf{C}^T(t) \\ \mathbf{A}(t) & \mathbf{0} & \mathbf{0} \\ (\mathbf{K}_1(t) - \mathbf{I})\mathbf{C}(t) & \mathbf{0} & \mathbf{I} - \mathbf{K}_2(t) \end{bmatrix}$$

$$\mathbf{M}(t) = \begin{bmatrix} \dot{\mathbf{S}}(t) & \dot{\mathbf{A}}^T(t) & \dot{\mathbf{C}}^T(t) \\ \dot{\mathbf{A}}(t) & \mathbf{0} & \mathbf{0} \\ (-\mathbf{I} + \mathbf{K}_1(t))\dot{\mathbf{C}}(t) & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$\mathbf{v}(t) = \begin{bmatrix} \dot{\mathbf{u}}_1(t) \\ \dot{\mathbf{u}}_2(t) \\ \dot{\mathbf{d}}(t) - \mathbf{K}_1(t)\dot{\mathbf{d}}(t) \end{bmatrix}$$

with $\mathbf{K}_1(t) := \Lambda(\mathbf{h}(t) \oslash \mathbf{n}(t))$, $\mathbf{K}_2(t) := \Lambda(\boldsymbol{\lambda}(t) \oslash \mathbf{n}(t))$, and $\mathbf{n}(t) := \sqrt{\mathbf{h}(t) \oslash \mathbf{h}(t) + \boldsymbol{\lambda}(t) \oslash \boldsymbol{\lambda}(t) + \boldsymbol{\delta}^+}$. In addition, the symbol $\oslash$ denotes the Hadamard division and $\Lambda(\cdot)$ represents creating a diagonal matrix using corresponding vector.

In-depth research studies have already found that the nonlinear AFs contribute greatly to the performance of ZNN models, especially their convergence speed. Traditional AFs can only offer exponential convergence rate, while SBP [15], [23] and some other AFs can achieve finite-time convergence time. To further enhance the performance of the PTCR-ZNN model (10), we propose a novel PPAF. The definition of the PPAF is as follows:

$$f(\varepsilon) = \begin{cases} \text{sgn}^{r_1}(\varepsilon), & |\varepsilon| > 1 \\ \frac{r_1}{r_2} \text{sgn}^{r_2}(\varepsilon), & |\varepsilon| \leq 1 \end{cases} \quad (11)$$

where $r_1 > 1$, $0 < r_2 < 1$, and the function $\text{sgn}^r(\cdot)$ is defined as

$$\text{sgn}^r(\varepsilon) = \begin{cases} |\varepsilon|^r, & \varepsilon \geq 0 \\ -|\varepsilon|^r, & \varepsilon < 0 \end{cases}.$$

Besides, in order to accelerate the convergence speed of the PTCR-ZNN model (10), both $\Psi_1(\cdot)$ and $\Psi_2(\cdot)$ use PPAF (11) in the following simulation part.

IV. CONVERGENCE ANALYSIS

In this section, we will prove that the PTCR-ZNN model (10) is globally stable and can converge to the optimal solution of the TVQPEI problem in a predefined time. Considering the i th ($\forall i \in 1, 2, \dots, n + m + p$) element of (9), we have the following subsystem:

$$\dot{\varepsilon}_i(t) = -\gamma_1 \psi_1(\varepsilon_i(t)) - \gamma_2 \psi_2 \left(\varepsilon_i(t) + \gamma_1 \int_0^t \psi_1(\varepsilon_i(\tau)) d\tau \right) \quad (12)$$

where $\varepsilon_i(t)$ denotes the i th element of $\boldsymbol{\varepsilon}(t)$, and $\psi_1(\cdot)$ and $\psi_2(\cdot)$ represent the i th element of $\Psi_1(\cdot)$ and $\Psi_2(\cdot)$, respectively. We introduce a new auxiliary function for analyzing (12) as

$$z_i(t) = \varepsilon_i(t) + \gamma_1 \int_0^t \psi_1(\varepsilon_i(\tau)) d\tau. \quad (13)$$

Then, seeking the derivative of (13) about time, we have

$$\dot{z}_i(t) = \dot{\varepsilon}_i(t) + \gamma_1 \psi_1(\varepsilon_i(t)). \quad (14)$$

Substituting (14) into (12), (12) can be reconstructed as $\dot{z}_i(t) = -\gamma_2 \psi_2(z_i(t))$, which is exactly the standard ZNN design formula for $z_i(t)$.

Theorem 1: The new design dynamic (9) is globally Lyapunov stable, meaning that $\boldsymbol{\varepsilon}(t)$ converges to $\mathbf{0}$ globally.

Proof: We come to prove that any i th subsystem of (9) is globally stable. A Lyapunov function candidate for (12) is defined as $L_i(t) = \frac{1}{2} k \varepsilon_i^2(t) + \frac{1}{2} z_i^2(t)$, where $k > 0$. Clearly, $L_i(t) > 0$ holds for any $\varepsilon_i(t) \neq 0$ or $z_i(t) \neq 0$, and $L_i(t) = 0$ come true if and only if $\varepsilon_i(t) = z_i(t) = 0$. Thus, the above Lyapunov function is positive definite. Calculating the time derivative of $L_i(t)$, we obtain

$$\begin{aligned} \frac{dL_i(t)}{dt} &= k \varepsilon_i(t) \dot{\varepsilon}_i(t) + z_i(t) \dot{z}_i(t) \\ &= k \varepsilon_i(t) [\dot{z}_i(t) - \gamma_1 \psi_1(\varepsilon_i(t))] - \gamma_2 z_i(t) \psi_2(z_i(t)) \\ &= -k \gamma_1 \varepsilon_i(t) \psi_1(\varepsilon_i(t)) - \gamma_2 z_i(t) \psi_2(z_i(t)) \\ &\quad - k \gamma_2 \varepsilon_i(t) \psi_2(z_i(t)). \end{aligned} \quad (15)$$

Let $L_0 = L_i(0) = k \varepsilon_i^2(0)/2 + z_i^2(0)/2$, and our first goal is proving that there exists k that makes $L_i(t) \leq 0$. Suppose there exists any time instant $t_a \in [0, +\infty)$ that satisfies $L_i(t_a) \leq L_0$. We have $k \varepsilon_i^2(t_a)/2 \leq L_0$ and $z_i^2(t_a) \leq L_0$, which leads to $0 \leq |\varepsilon_i(t_a)| \leq \sqrt{2L_0/k}$ and $0 \leq |z_i(t_a)| \leq \sqrt{2L_0}$.

Let us define two domains as follows:

$$\begin{aligned} D_1 &:= \{z_i(t) \in R, \quad |z_i(t)| \leq \sqrt{2L_0}\} \\ D_2 &:= \{\varepsilon_i(t) \in R, \quad |\varepsilon_i(t)| \leq \sqrt{2L_0/k}\}. \end{aligned}$$

We use the mean value theorem in D_1 and obtain

$$\psi_2(z_i(t_a)) - \psi_2(0) = (z_i(t_a) - 0) \frac{\partial \psi_2(z_i(\vartheta))}{\partial z_i} \Big|_{z_i(\vartheta) \in D_1}.$$

Because PPAF (11) satisfies $\psi_2(0) = 0$ and $\partial \psi_2(z_i(t))/\partial z_i > 0$, we can derive from the above equation that $|\psi_2(z_i(t_a))| \leq W_0 |z_i(t_a)|$, where $W_0 = \max\{\partial \psi_2(z_i(t))/\partial z_i\}_{z_i(t) \in D_1} > 0$. Therefore, we have

$$\begin{aligned} |\varepsilon_i(t_a) \psi_2(z_i(t_a))| &\leq |\varepsilon_i(t_a)| \cdot |\psi_2(z_i(t_a))| \\ &\leq W_0 |\varepsilon_i(t_a)| \cdot |z_i(t_a)|. \end{aligned}$$

Combining (15) with above inequality, we have

$$\begin{aligned} \dot{L}_i(t_a) &= -k \gamma_2 \varepsilon_i(t_a) \psi_2(z_i(t_a)) - k \gamma_1 \varepsilon_i(t) \psi_1(\varepsilon_i(t_a)) \\ &\quad - \gamma_2 z_i(t_a) \psi_2(z_i(t_a)) \\ &\leq k \gamma_2 |\varepsilon_i(t_a) \psi_2(z_i(t_a))| - k \gamma_1 \varepsilon_i(t_a) \psi_1(\varepsilon_i(t_a)) \\ &\quad - \gamma_2 z_i(t_a) \psi_2(z_i(t_a)) \\ &\leq k \gamma_2 W_0 |\varepsilon_i(t_a)| |z_i(t_a)| - k \gamma_1 W_1 \varepsilon_i^2(t_a) \\ &\quad - \gamma_2 W_2 z_i^2(t_a) \\ &= -k \left(\sqrt{\gamma_1 W_1} |\varepsilon_i(t_a)| - \gamma_2 W_0 |z_i(t_a)| / \left(2\sqrt{\gamma_1 W_1} \right) \right)^2 \\ &\quad - k \left(\gamma_2 W_2 / k - \gamma_2^2 W_0^2 / (4\gamma_1 W_1) \right) z_i^2(t_a) \end{aligned} \quad (16)$$

where $W_1 = \min\{\partial\psi_1(\varepsilon_i(t))/\partial\varepsilon_i\}_{|\varepsilon_i(t) \in D_2} > 0$ and $W_2 = \min\{\partial\psi_2(z_i(t))/\partial z_i\}_{|z_i(t) \in D_1} > 0$, which are also obtained using the mean value theorem like calculating W_0 . From (16), we know that $\dot{L}_i(t_a) \leq 0$ is guaranteed as long as k is selected such that

$$\frac{\gamma_2 W_2}{k} - \frac{\gamma_2^2 W_0^2}{4\gamma_1 W_1} \geq 0, \quad k > 0 \Leftrightarrow 0 < k \leq \frac{4\gamma_1 W_1 W_2}{\gamma_2 W_0^2}.$$

After $\dot{L}_i(t_a) \leq 0$, $L_i(t)$ will not increase for $t \geq t_a$ since $L_i(t) \leq L_0$ is always true. It is worth noting that $t_a = 0$ exists for all circumstances; the proof of $\dot{L}_i(t) \leq 0$ is complete. Therefore, (9) is globally stable in the sense of Lyapunov. ■

Theorem 2: Using PPAF (11) in $\Psi(\cdot)$, the normal ZNN model (8) converge to 0 in predefined time $T(\gamma, r_1, r_2)$:

$$T(\gamma, r_1, r_2) \leq \frac{1}{\gamma(r_1 - 1)} + \frac{r_2}{r_1} \cdot \frac{1}{\gamma(1 - r_2)}$$

where γ is defined in (8) and r_1 and r_2 are defined in (11).

Proof: Take the i th ($\forall i \in 1, \dots, n + m + p$) subsystem of the ZNN model (8) into consideration

$$\dot{\varepsilon}_i(t) = -\gamma\psi(\varepsilon_i(t)) \quad (17)$$

where $\psi(\cdot)$ is the PPAF, and $\varepsilon_0 = \varepsilon_i(0)$ is unknown. Let us first consider the condition that $\varepsilon_0 \geq 0$.

In the case of $\varepsilon_0 > 1$, $\varepsilon_i(t)$ should converge in two stages: from ε_0 to 1 and from 1 to 0. Assuming that the first convergence stage takes time t_1 , (17) becomes

$$\varepsilon_i^{-r_1} d\varepsilon_i = -\gamma dt.$$

Integrating the last equation of above equations from $t = 0$ to $t = t_1$, we obtain

$$\int_{\varepsilon_0}^1 \varepsilon_i^{-r_1} d\varepsilon_i = -\gamma \int_0^{t_1} dt.$$

Calculating the above integral equation gives $t_1 = (1 - \varepsilon_0^{1-r_1})/[\gamma(r_1 - 1)] \leq 1/[\gamma(r_1 - 1)]$. If the second converge stage takes time t_2 , (17) becomes

$$\varepsilon_i^{-r_2} d\varepsilon_i = -\gamma \frac{r_1}{r_2} dt.$$

Similarly, we integrate above equality from $t = t_1$ to $t = t_1 + t_2$ as follows:

$$\int_1^0 \varepsilon_i^{-r_2} d\varepsilon_i = -\gamma \frac{r_1}{r_2} \int_{t_1}^{t_1+t_2} dt.$$

Solving the above integral equation, we obtain $t_2 = r_2/[r_1\gamma(1 - r_2)]$. Therefore, when $\varepsilon_0 > 1$, $\varepsilon_i(t)$ converges to 0 in predefined time $t_1 + t_2$. In other case of $0 \leq \varepsilon_0 \leq 1$, the subsystem falls within the second convergence stage, which still satisfies the time upper bound $t_1 + t_2$.

As for the case of $\varepsilon_0 < 0$, we can get the same convergence time upper bound $t_1 + t_2$ using the aforementioned technique. Thus, all subsystems of (8) converge to zero in predefined time $T(\gamma, r_1, r_2) = t_1 + t_2$. In other words, the ZNN model (8) converges to 0 in predefined time $T(\gamma, r_1, r_2)$. ■

Theorem 3: The PTCR-ZNN model (9) will globally converge to 0 in predefined time $T_p = T(\gamma_1, r_1, r_2) + T(\gamma_2, r_1, r_2)$:

$$T_p \leq \frac{\gamma_1 + \gamma_2}{\gamma_1 \gamma_2 (r_1 - 1)} + \frac{r_2}{r_1} \cdot \frac{\gamma_1 + \gamma_2}{\gamma_1 \gamma_2 (1 - r_2)} \quad (18)$$

using the same PPAFs (11) in $\Psi_1(\cdot)$ and $\Psi_2(\cdot)$, with γ_1 and γ_2 defined in (9) and r_1 and r_2 defined in (11).

Proof: With the auxiliary function $z_i(t)$ defined in (13), we know that the i th subsystem of the PTCR-ZNN model (12) can be reformulated into the dynamic (14), which is equivalent to the ordinary ZNN model (17). Therefore, we conclude from Theorem 2 that $z_i(t)$ must converge to 0 in predefined time $t_3 = T(\gamma_2, r_1, r_2)$. Then, $z_i(t) = 0$ will hold after time period t_3 , meaning that $\dot{z}_i(t) = 0$ is satisfied for $t \geq t_3$. We thus have $\dot{\varepsilon}_i(t) = -\gamma_1 \psi_1(\varepsilon_i(t))$, $t \geq t_3$ according to (14). Evidently, the above equality also conforms to the normal ZNN model (17). Hence, $\varepsilon_i(t)$ decreases to zero after another time period $t_4 = T(\gamma_1, r_1, r_2)$. Finally, the PTCR-ZNN model (9) with its all subsystems converges to zero in predefined time $T_p = t_3 + t_4 = T(\gamma_1, r_1, r_2) + T(\gamma_2, r_1, r_2)$. The proof of Theorem 3 is now complete. ■

V. ROBUSTNESS ANALYSIS

The superiority of the PTCR-ZNN model (9) lies not only in the fast and controllable convergence speed under ideal conditions, but also in its powerful noise suppression ability. In this section, we consider the additive noise-polluted PTCR-ZNN model

$$\begin{aligned} \dot{\varepsilon}(t) = & -\gamma_1 \Psi_1(\varepsilon(t)) - \gamma_2 \Psi_2 \left(\varepsilon(t) + \gamma_1 \int_0^t \Psi_1(\varepsilon(\tau)) d\tau \right) \\ & + \xi(t) \end{aligned} \quad (19)$$

where $\xi(t) \in \mathbb{R}^{n+m+p}$ is unknown time-varying additive noise. The i th ($\forall i \in 1, \dots, n + m + p$) subsystem of (19) can be expressed as

$$\begin{aligned} \dot{\varepsilon}_i(t) = & -\gamma_1 \psi_1(\varepsilon_i(t)) - \gamma_2 \psi_2 \left(\varepsilon_i(t) + \gamma_1 \int_0^t \psi_1(\varepsilon_i(\tau)) d\tau \right) \\ & + \xi_i(t) \end{aligned} \quad (20)$$

where $\xi_i(t)$ is the i th element of $\xi(t)$. Similarly, the conventional ZNN model polluted by additive noise is

$$\dot{\varepsilon}(t) = -\gamma \Psi(\varepsilon(t)) + \xi(t). \quad (21)$$

Theorem 4: If the time derivative of $\xi_i(t)$ in (20) exists and is bounded for any time $t \geq 0$, the residual error $\|\varepsilon(t)\|_2$ of the polluted PTCR-ZNN model (19) tends to converge to the bounded interval $[0, l|\dot{\xi}_{\max}(t)|/(\gamma_1 \gamma_2 \varrho)]$ when $t \rightarrow +\infty$, where $l = n + m + p$, $|\dot{\xi}_{\max}(t)| = \max\{|\dot{\xi}_i(t)|\}$, and $\varrho = \min\{\varrho_i\}$ with $\varrho_i = |\psi_1(\varepsilon_i(t))|/|\varepsilon_i(t)| \geq 1$.

Proof: Defining a function $g_i(t) = z_i^2(t)/2 = |z_i(t)|^2/2$ for $z_i(t)$ in (13), the perturbed neural dynamic (20) is equivalent to $\dot{z}_i(t) = -\gamma_2 \psi_2(z_i(t)) + \xi_i(t)$. Following this equation, the time derivative of $g_i(t)$ can be calculated as:

$$\dot{g}_i(t) = \dot{z}_i(t)z_i(t) = (-\gamma_2 \psi_2(z_i(t)) + \xi_i(t))z_i(t).$$

In view of $\psi_2(z_i(t))z_i(t) \geq 0$, we have that $|z_i(t)|$ may increase only if $\xi_i(t)z_i(t) > 0$. If $|z_i(t)|$ is increasing under such condition, since $\xi_i(t)$ and $z_i(t)$ have the same sign, $-\gamma_2\psi_2(z_i(t)) + \xi_i(t)$ is decreasing in the same time. However, the increasing of $|z_i(t)|$ stops when $-\gamma_2\psi_2(z_i(t)) + \xi_i(t) = 0$, i.e., $\dot{g}_i(t) = 0$. Therefore, with $t \rightarrow +\infty$, we have

$$0 \leq |z_i(t)| \leq |\psi_2^{-1}(\xi_i(t)/\gamma_2)| - |\xi_i(t)/\gamma_2| \leq z_i(t) \leq |\xi_i(t)/\gamma_2| \quad (22)$$

where $\psi_2^{-1}(\cdot)$ denotes the inverse function of $\psi_2(\cdot)$, and we have $|\psi_2(x)| \geq |x|$, meaning that $|\psi_2^{-1}(x)| \leq |x|$ for PPAF (11).

Now, we continue to prove that with $t \rightarrow +\infty$, $\gamma_1\psi_1(\varepsilon_i(t))$ converges to the bounded interval $[-|\dot{\xi}_i(t)/\gamma_2|, |\dot{\xi}_i(t)/\gamma_2|]$. We suppose there exists a time instant $t_a > 0$ that makes $E_1 = z_i(t_a)$ and $\gamma_1\psi_1(\varepsilon_i(t)) \notin [-|\dot{\xi}_i(t)/\gamma_2|, |\dot{\xi}_i(t)/\gamma_2|]$. In order to use the contradiction technique, we suppose that $\gamma_1\psi_1(\varepsilon_i(t)) > |\dot{\xi}_i(t)/\gamma_2| > 0$ holds true for $t \in [t_a, t_a + t_b]$. Thus, after a time period t_b , we get $E_2 = z_i(t_a + t_b)$. The difference between E_1 and E_2 in light of (13) is

$$\begin{aligned} E_2 - E_1 &= (\varepsilon_i(t_a + t_b) - \varepsilon_i(t_a)) + \gamma_1 \int_0^{t_a+t_b} \psi_1(\varepsilon_i(\tau)) d\tau \\ &\quad - \gamma_1 \int_0^{t_a} \psi_1(\varepsilon_i(\tau)) d\tau \\ &= (\varepsilon_i(t_a + t_b) - \varepsilon_i(t_a)) + \gamma_1 \int_{t_a}^{t_a+t_b} \psi_1(\varepsilon_i(\tau)) d\tau \\ &\geq \gamma_1 \int_{t_a}^{t_a+t_b} \psi_1(\varepsilon_i(\tau)) d\tau - \varepsilon_i(t_a). \end{aligned} \quad (23)$$

It should be noted that we can obtain

$$\begin{aligned} &|\xi_i(t_a + t_b)/\gamma_2| - E_1 \\ &\leq |\xi_i(t_a)/\gamma_2| + |\xi_i(t_a + t_b)/\gamma_2 - \xi_i(t_a)/\gamma_2| - E_1 \\ &= |\xi_i(t_a)/\gamma_2| + \left| \int_{t_a}^{t_a+t_b} (\dot{\xi}_i(\tau)/\gamma_2) d\tau \right| - E_1 \\ &\leq |\xi_i(t_a)/\gamma_2| + \int_{t_a}^{t_a+t_b} |\dot{\xi}_i(\tau)/\gamma_2| d\tau - E_1. \end{aligned} \quad (24)$$

Computing (23) and (24) leads to

$$\begin{aligned} E_2 - |\xi_i(t_a + t_b)/\gamma_2| &\geq \int_{t_a}^{t_a+t_b} (\gamma_1\psi_1(\varepsilon_i(\tau)) - |\dot{\xi}_i(\tau)/\gamma_2|) d\tau \\ &\quad - (|\xi_i(t_a)/\gamma_2| + \varepsilon_i(t_a) - E_1). \end{aligned}$$

With $t_a \rightarrow +\infty$, (22) tells us that $0 \geq E_2 - |\xi_i(t_a + t_b)/\gamma_2|$. It follows from the above inequality that

$$\begin{aligned} &\int_{t_a}^{t_a+t_b} (\gamma_1\psi_1(\varepsilon_i(\tau)) - |\dot{\xi}_i(\tau)/\gamma_2|) d\tau \\ &\leq |\xi_i(t_a)/\gamma_2| + \varepsilon_i(t_a) - E_1. \end{aligned} \quad (25)$$

For the left-hand side of (25), $\gamma_1\psi_1(\varepsilon_i(t)) - |\dot{\xi}_i(t)/\gamma_2| \geq 0$, $t \in [t_a, t_a + t_b]$ according to the previous assumption. Thus, the left-hand side of (25) always increases with t_b increasing.

Considering that the right-hand side of (25) is a finite fixed value, we are able to conclude that $\gamma_1\psi_1(\varepsilon_i(t)) - |\dot{\xi}_i(t)/\gamma_2| \rightarrow 0$ by the contradiction of (25) while $t_b \rightarrow +\infty$. Under another assumption when $\gamma_1\psi_1(\varepsilon_i(t)) < -|\dot{\xi}_i(t)/\gamma_2| < 0$, we can similarly have the conclusion that $\gamma_1\psi_1(\varepsilon_i(t)) + |\dot{\xi}_i(t)/\gamma_2| \rightarrow 0$.

The above conclusion implies that $\gamma_1\psi_1(\varepsilon_i(t))$ converges to the interval $[-|\dot{\xi}_i(t)/\gamma_2|, |\dot{\xi}_i(t)/\gamma_2|]$ under all circumstances, suggesting that $\varepsilon_i(t)$ will converge to $[-|\dot{\xi}_i(t)/(\gamma_1\gamma_2\varrho_i)|, |\dot{\xi}_i(t)/(\gamma_1\gamma_2\varrho_i)|]$. Moreover, for the residual error $\|\varepsilon(t)\|_2$, we get $0 \leq \|\varepsilon(t)\|_2 = \sqrt{\sum_{i=1}^l \varepsilon_i^2(t)} \leq l|\varepsilon_{\max}(t)|$, where $|\varepsilon_{\max}(t)|$ denotes the largest element among $|\varepsilon_i(t)|$, $i \in 1, \dots, l$. Eventually, $\|\varepsilon(t)\|_2$ converges to the bounded interval

$$\|\varepsilon(t)\|_2 \in [0, l|\dot{\xi}_{\max}(t)|/(\gamma_1\gamma_2\varrho)]. \quad (26)$$

The proof is now complete. $\blacksquare$

Remark 1: Unlike convergence time, perturbations in realistic applications are usually unknown, and we can only try to eliminate their negative impacts. For this reason, we provide Theorem 4 for the PTCR-ZNN model (9) to help us evaluate and control the perturbed PTCR-ZNN model's residual error under nonstationary unknown noises, instead of studying only limited types of noises like in [21] and [26]. We learn from Theorem 4 that, as long as the time derivative of unknown noises is bounded, steady error $\|\varepsilon(t)\|_2$ is bounded for model (19) regardless of the size of noises. In addition, larger γ_1 , γ_2 , and ϱ in the AF lead to smaller residual error.

VI. SIMULATION VALIDATION AND COMPARISON

A. Qualitative Experiments

For validating previous theoretical results as well as further illustrating PTCR-ZNN's convergence and robustness advantages over the conventional ZNN model (8) for TVQP problems, a series of simulations are conducted in this section.

For the purpose of testing the predefined-time convergence performance of the PTCR-ZNN model (10) on solving TVQP problems, both TVQPEI (1)–(3) and TVQPE (4) have been tested. It is worth noting that all elements in δ^+ are set to be 10^{-8} , and the simulation duration is 10 s or 1 s in this section. The TVQPEI example is as follows:

$$\begin{aligned} &\min (\sin(t)/4 + 1)/2 \cdot x_1^2(t) + (\cos(t)/4 + 1)/2 \cdot x_2^2(t) \\ &\quad \cos(t)x_1(t)x_2(t)/2 + \sin(3t)x_1(t) + \cos(3t)x_2(t) \end{aligned} \quad (27)$$

$$\text{s.t. } \sin(4t)x_1(t) + \cos(4t)x_2(t) = \cos(2t), \quad (28)$$

$$-1.2 \leq x_1(t), x_2(t) \leq 1.2 \quad (29)$$

where $x_1(0)$ and $x_2(0)$ are randomly generated in $[0, 1]$. Other parameters are $\gamma_1 = \gamma_2 = 1$ and $r_1 = 2, r_2 = 0.5$ in the PTCR-ZNN model (10). The results of using PTCR-ZNN model (10) to solve (27)–(29) are presented in Figs. 1(a) and 2(a). From Theorem 3, we conclude that the residual error $\|\varepsilon(t)\|_2$ should converge to 0 in predefined time $T_p = 2 * T(1, 2, 0.5) \leq 3$ s, and it can be seen in Fig. 2(a) that $\|\varepsilon(t)\|_2$ decreases to 0 within 2 s in the simulation. Clearly, the PTCR-ZNN model (10) can handle the TVQPEI problem efficiently and obtain the optimal solution in a predefined time. For the TVQPE problem

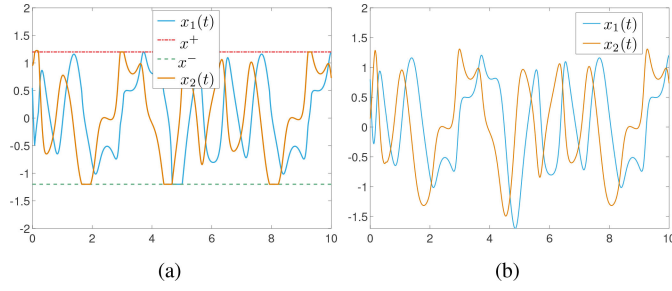


Fig. 1. State solutions computed by the PTCR-ZNN model (9) on solving different TVQP problems. (a) TVQPEI (27)–(29). (b) TVQPE (30), (31).

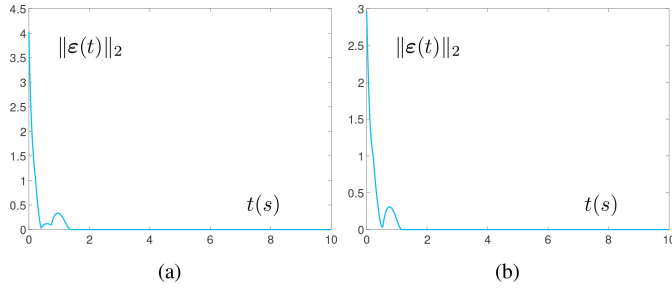


Fig. 2. Residual error $\|\varepsilon(t)\|_2$ generated by the PTCR-ZNN model (9) on solving different TVQP problems. (a) TVQPEI (27)–(29). (b) TVQPE (30), (31).

simulation example, we consider

$$\begin{aligned} \min & (\sin(t)/4 + 1)/2 \cdot x_1^2(t) + (\cos(t)/4 + 1)/2 \cdot x_2^2(t) \\ & \cos(t)x_1(t)x_2(t)/2 + \sin(3t)x_1(t) + \cos(3t)x_2(t) \quad (30) \\ \text{s.t.} & \sin(4t)x_1(t) + \cos(4t)x_2(t) = \cos(2t). \quad (31) \end{aligned}$$

From Section II-A, we know that the TVQPE problem can be easily reconstructed into the TVQPEI problem by adopting $\mathbf{C}(t) = [I_n, -I_n]^T$ and $\mathbf{d}(t) = [\infty_n^T, \infty_n^T]^T$ in (3). Therefore, we use real number 10^8 to denote the positive infinity, and other parameters are the same as in the above TVQPEI problem. Such experiment results can be found in Figs. 1(b) and 2(b). Fig. 2(b) shows that the predefined-time convergence of the PTCR-ZNN model (10) is validated again, in which $\|\varepsilon(t)\|_2$ drops to 0 within $2 \leq 3$ s. Furthermore, comparing the state solutions of TVQPE (30), (31) in Fig. 1(b) with that of TVQPEI (27), (28) in Fig. 1(a), they strictly subject to inequality constraints and coincide with each other when the optimal solutions of (27)/(30) falls inside the feasible domain.

Consider the convergence time upper bound (18) in Theorem 3; the relationship between T_p and four parameters γ_1, γ_2, r_1 , and r_2 is so complicated that we cannot validate it using only above conducted experiments. Hence, we shorten the designated upper bound T_p to 1 s and choose four different groups of design parameters to meet this restriction. In the first two groups, $r_1 = 2$ and $r_2 = 0.5$ are kept unchanged, and we set $\gamma_1 = 2, \gamma_2 = 6$ or $\gamma_1 = 6, \gamma_2 = 2$. For the last two groups, we keep $\gamma_1 = \gamma_2 = 1$ unchanged and set $r_1 = 4, r_2 = 0.4$ or $r_1 = 6, r_2 = 0.6$ instead.

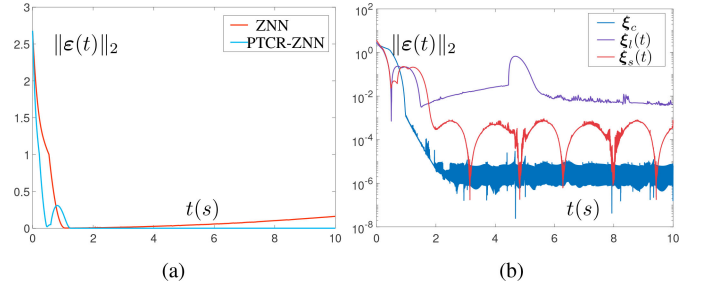


Fig. 3. Comparisons of residual error $\|\varepsilon(t)\|_2$ produced by different models with different noises. (a) By perturbed ZNN (21) and perturbed PTCR-ZNN (19) with $0.1\xi_l(t)$. (b) By perturbed PTCR-ZNN with $\xi_c, \xi_l(t)$ and $\xi_s(t)$.

Theorem 4 indicates that the stable computation error $\|\varepsilon(t)\|_2$ of the perturbed PTCR-ZNN model (19) is bounded and controllable even facing with time-varying additive noises. Thus, we have selected three representative noises: constant noise ξ_c , linear noise $\xi_l(t)$, and sinusoidal noise $\xi_s(t)$. All elements of $\xi_c, \xi_l(t)$, and $\xi_s(t)$ are 3, t , and $\sin(t)$, respectively. A linear noise $0.1\xi_l(t)$ is first used to demonstrate the superior robustness of the perturbed PTCR-ZNN model (19) against the normal perturbed ZNN model (21) for solving (27)–(29), where the ZNN uses the same PPAF as in PTCR-ZNN (19) with $\gamma = \gamma_1 = \gamma_2 = 1, r_1 = 2$, and $r_2 = 0.5$. Fig. 3(a) depicts the results of this experiment, which shows that even with small noise pollution, the ordinary ZNN model performs much worse than PTCR-ZNN model (19). It is reasonable to assume that $\|\varepsilon(t)\|_2$ of the ZNN model will continue to rise above 0.5 after 10 s, while the one of PTCR-ZNN model (19) is always kept at a low and stable level.

In order to further verify the conclusion of Theorem 4, we apply all three larger noise $\xi_c, \xi_l(t), \xi_s(t)$ to the PTCR-ZNN model (19) while keeping other parameters unchanged. The results of solving (27)–(29) by the PTCR-ZNN model (19) are plotted in Fig. 3(b). Following the properties of those noises' time derivative, we can see in Fig. 3(b) that: for ξ_c , the stable residual error $\|\varepsilon(t)\|_2$ is nearly zero, where the precision of this simulation is at level of 10^{-6} ; for $\xi_l(t)$, $\|\varepsilon(t)\|_2$ is slightly higher but still very small and stable (at the level of 10^{-2}); and for $\xi_s(t)$, due to the change of its time-varying time derivative, the stable computation error $\|\varepsilon(t)\|_2$ is also changing, but always below that of the linear noise case. These inversely validate the correctness of Theorem 4.

B. Quantitative Experiments

Through the previous analysis, we have derived precise quantitative upper bound formulas for predefined convergence time (18) and stable residual error (26), and it is necessary to verify these formulas through more case studies and quantitative comparisons. Therefore, another example of TVQPEI problem called TVQPEI case 2 (TVQPEI-C2) is introduced as follows:

$$\mathbf{S} = \begin{bmatrix} 2 * \sin(t) + 2 & \exp(\sin(t))/2 \\ \exp(\cos(t))/2 & \cos(t) + 3 \end{bmatrix}$$

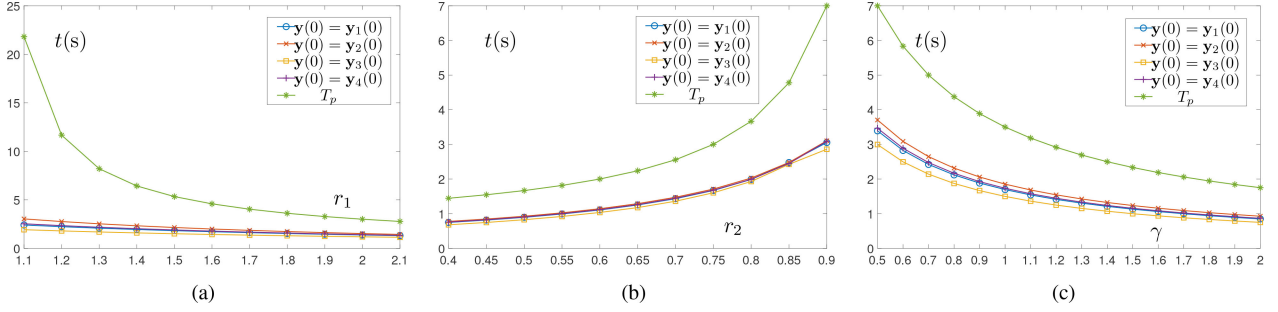


Fig. 4. Convergence time of the PTCR-ZNN model (10) solving the TVQPEI-C2 problem starting from different initial conditions $\mathbf{y}(0)$ with different model variable settings. T_p is the predefined convergence time bound by (18). (a) $r_2 = 0.5$ and $\gamma_1 = \gamma_2 = 1$ while changing r_1 . (b) $r_1 = 3$ and $\gamma_1 = \gamma_2 = 1$ while changing r_2 . (c) $r_1 = 2$, $r_2 = 0.6$, and $\gamma_1 = \gamma_2$ while changing γ_1 and γ_2 .

$\mathbf{p}(t) = [\cos(2t), \sin(t+1)]^T$, $\mathbf{A}(t) = [2 * \sin^2(t), \log(4 + \cos(t))]$, $\mathbf{b} = \cos(t) + \sin(3t)$, $\mathbf{C}(t) = [I_2, -I_2]^T$, $\mathbf{d}(t) = [\sin(t) + 2, \sin(t) + 2, \sin(t) + 2, \sin(t) + 2]^T$, and $\mathbf{x}(t) \in \mathbb{R}^2$.

Compared to the TVQPEI problem used in Section VI-A, TVQPEI-C2 has added more nonlinearity, including exponential, power, logarithmic functions and the time-varying inequalities. Thus, TVQPEI-C2 can better represent TVQPEI problems that we encounter in the real world. For controlling variables in the following experiments for solving the TVQPEI-C2 problem, we have generated four different initial values $\mathbf{y}(0) \in \mathbb{R}^4$ for the PTCR-ZNN model (10) in advance. They are denoted by $\mathbf{y}_1(0)$, $\mathbf{y}_2(0)$, $\mathbf{y}_3(0)$, and $\mathbf{y}_4(0)$ of which elements are randomly generated in interval $[-1, 1]$.

For TVQPEI-C2, let us first verify the predefined convergence time upper bound formula (18) with quantitative experiments. Note that due to the simulation precision limit, we have shown in Fig. 3 that the neural network can be deemed to converge to zero when its residual error approaches to about 10^{-6} . Therefore, the convergence time t_c in the following experiments is defined as the earliest time instant when $\|\varepsilon(t_c)\|_2 \leq 10^{-5}$. According to formula (18), the predefined convergence time bound T_p of the PTCR-ZNN model (9) is relevant to r_1 , r_2 , γ_1 , and γ_2 . Thus, we have conducted three experiments, which validate roles of these variables, respectively. In these three experiments, the PTCR-ZNN model (10) is used to solve the TVQPEI-C2 problem. In the first experiment, we fix the last three model variables as $r_2 = 0.5$, $\gamma_1 = \gamma_2 = 1$ and change the first as $r_1 = 1.1, 1.2, 1.3, \dots, 2.1$. Starting from different $\mathbf{y}(0)$, results and corresponding T_p of the PTCR-ZNN model (10) are plotted in Fig. 4(a). In the second experiment, we fix three model variables as $r_1 = 3$, $\gamma_1 = \gamma_2 = 1$ and change the other model variable as $r_2 = 0.4, 0.45, 0.5, \dots, 0.9$. Starting from different $\mathbf{y}(0)$, results of the PTCR-ZNN model (10) together with corresponding T_p are shown in Fig. 4(b). As for the third experiment, we keep $r_1 = 2$, $r_2 = 0.6$, $\gamma_1 = \gamma_2$ and change γ_1, γ_2 in $0.5, 0.6, 0.7, \dots, 2$ with results illustrated in Fig. 4(c). In Fig. 4(a)–(c), the convergence time of all simulations is clearly less than T_p predicted by formula (18). We also find that even if the convergence time may be slightly affected by the initial value $\mathbf{y}(0)$, it still strictly conforms to our theoretical prediction. Another important observation of Fig. 4(a)–(c) is that

the changing direction of convergence time strictly conforms to the variable relationships reflected in formula (18) as well, which verifies the PTCR-ZNN model (10)'s tunable predefined-time convergence again.

Now, after verifying T_p in convergence time bound (18), we would like to quantitatively examine PTCR-ZNN model (10)'s stable residual error under time-varying noises, which should coincide with bound (26). According to the error bound formula (26) and Fig. 3(b), we conclude that $\|\varepsilon(t)\|_2$ may be nonstationary due to the time-varying noise. Thus, in the following experiments, the maximum steady-state residual error (MSSRE) is introduced for evaluation, and the MSSRE is defined as $\max\{\|\varepsilon(t)\|_2\} \forall t \in [7, 10]$ s (the duration of simulation experiments are 10 s). Moreover, a new type of time-varying noise is designed as $\xi_c(t, a_n) = \cos(a_n t)$, $a_n \in \mathbb{R}^+$, of which the maximum value will not change with a_n but its time derivative $\dot{\xi}_c(t, a_n) = -a_n * \sin(a_n t)$ will. To validate the relationship between $\|\varepsilon(t)\|_2$ and $\dot{\xi}_{\max}(t)$ in inequality (26), we set $r_1 = 4$, $r_2 = 0.6$, $\gamma_1 = \gamma_2 = 1$, $\xi(t) = \xi_c(a_n, t)$ and change $a_n = 1, 1.5, 2, \dots, 10$. The MSSREs of the perturbed PTCR-ZNN model (19) when solving TVQPEI-C2 problem are presented in Fig. 5(a), where the linear relationship between MSSRE ($\|\varepsilon(t)\|_2$) and a_n ($\dot{\xi}_{\max}(t)$) is well illustrated. We also find that the initial condition of $\mathbf{y}(0)$ has a slight influence on the MSSRE. If we assume $\varrho = 4 \times 10^3$ with known $l = n + m + p = 7$, MSSRE upper bounds (MSSRE_{pred}) predicted by (26) coincide with actual MSSREs in Fig. 5(a) well, meaning that noise resistance of the PTCR-ZNN model (19) can be much stronger than the worst case when $\varrho = 1$. Next, we want to validate how γ_1 and γ_2 contribute in bound (26). We have $r_1 = 2$, $r_2 = 0.6$, $\mathbf{y}(0) = \mathbf{y}_1(0)$, and $a_n = 10$. Changing γ_1 and γ_2 individually and keeping other settings the same as previous noise simulations, the results are shown in Fig. 5(b). As can be observed in Fig. 5(b), the MSSREs of the noise-perturbed PTCR-ZNN model (19) always decrease with the increasing of γ_1 or γ_2 . Through this, it can also be verified that their decreasing curves match the fractional function about γ_1 and γ_2 revealed in bound (26). Finally, based on the bound of $\|\varepsilon(t)\|_2$ (26) and $\varrho = \min\{|\psi(\varepsilon_i(t))|/|\varepsilon_i(t)|\}$, we conjecture that r_1 and r_2 in PPAF (11) may affect the MSSRE as well by affecting ϱ . Thus, we keep $\gamma_1 = \gamma_2 = 1$, $\xi(t) = \xi_c(t, 10)$ unchanged and change r_1 and r_2 individually, and the

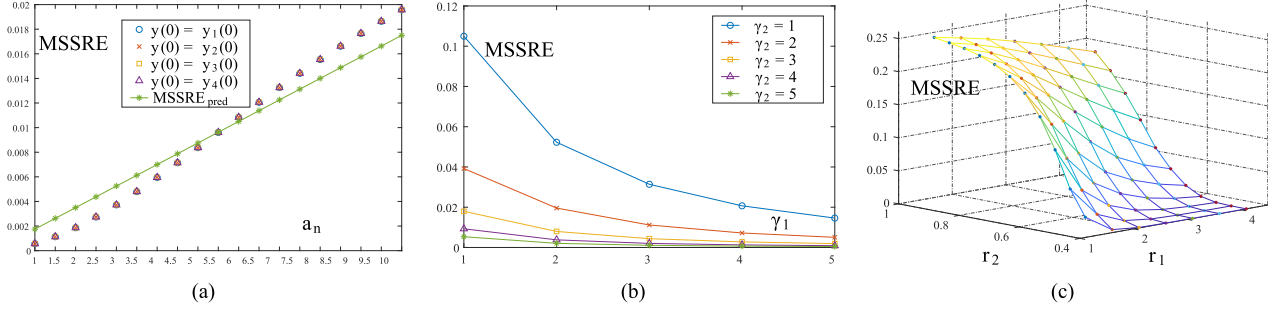


Fig. 5. MSSRE of $\xi_c(t, a_n)$ noise perturbed PTCR-ZNN model (21) solving the TVQPEI-C2 problem with different model variable settings, where $MSSRE_{pred}$ is the predicted MSSRE bound by inequality (26) when $\varrho = 4 \times 10^3$. (a) $r_1 = 4, r_2 = 0.6$, and $\gamma_1 = \gamma_2 = 1$ while changing a_n and $y(0)$. (b) $r_1 = 2, r_2 = 0.6, y(0) = y_1(0)$, and $a_n = 10$ while changing γ_1 and γ_2 . (c) $\gamma_1 = \gamma_2 = 1$, and $a_n = 10$ while changing r_1 and r_2 .

noise-perturbed PTCR-ZNN model (19) is used to solve TVQPEI-C2 with MSSREs plotted in Fig. 5(c). From Fig. 5(c), the MSSREs always drop with the increasing of r_1 or decreasing of r_2 , meaning that the MSSRE will drop if we adjust AF to increase ϱ .

Guided from above quantitative comparison experiments, we now conclude that our PTCR-ZNN model (10) can enhance convergence and noise suppression simultaneously by making r_1, γ_1 , and γ_2 as big as possible and r_2 as small as possible.

VII. APPLICATION TO IMAGE FUSION

In this section, the PTCR-ZNN model (10) is applied to solve the image fusion problem, which receives several noisy images for the same scene and fuses them into a single image. The resultant image should have higher quality than noisy images.

Suppose we have S different grayscale pictures with size of $I \times J$ for the same scene, which can be taken from different or the same sensor. The normalized pixels of those images form S matrices denoted by $\mathbf{P}_s(i, j)$, where $s = 1, 2, \dots, S; i = 1, 2, \dots, I; j = 1, 2, \dots, J$. Then, we row-wisely vectorize $\mathbf{P}_s$ into the s th column of $\mathbf{Z} \in \mathbb{R}^{L \times S}$, where $L = I \times J$. After defining $\mathbf{Z}_s$ as the s th column of $\mathbf{Z}$, we have $\mathbf{Z}_s((i-1)J + j) = \mathbf{P}_s(i, j)$. Considering pixels in $\mathbf{Z}$ that come from the same picture $\mathbf{P}_s$, we assume that they satisfy decompose equation $\mathbf{Z}_s = \alpha \circ \mathbf{p}_s + \mathcal{N}_s$, where $\alpha \in \mathbb{R}^L$ denotes a scaling vector, $\mathbf{p}_s \in \mathbb{R}^L$ is obtained by row-wisely vectoring $\mathbf{P}_s$ and $s = 1, \dots, S$. In addition, $\mathcal{N}_s \in \mathbb{R}^L$ stands for zero-mean Gaussian noise randomly produced during imaging. Under the above decompose equation, the best fused image is derived from

$$\mathbf{z}^* = \sum_{s=1}^S \mathbf{Z}_s \mathbf{x}^*(s) = \mathbf{Z} \mathbf{x}^*$$

where $\mathbf{x}^*$ is the optimal solution of the following QP problem [1], [2]:

$$\min \mathbf{x}^T \mathbf{G} \mathbf{x} / 2 \quad (32)$$

$$\text{s.t. } \mathbf{a}^T \mathbf{x} = 1, \mathbf{0} \leq \mathbf{x} \quad (33)$$

where $\mathbf{a} = [1, 1, \dots, 1]^T \in \mathbb{R}^S$ and $\mathbf{G} = (\sum_{l=1}^L \mathbf{r}_l \mathbf{r}_l^T) / L \in \mathbb{R}^{S \times S}$, with $\mathbf{r}_l^T \in \mathbb{R}^{1 \times S}$ representing the l th row of $\mathbf{Z}$. By transforming (32) and (33) into the standard QP problem, the

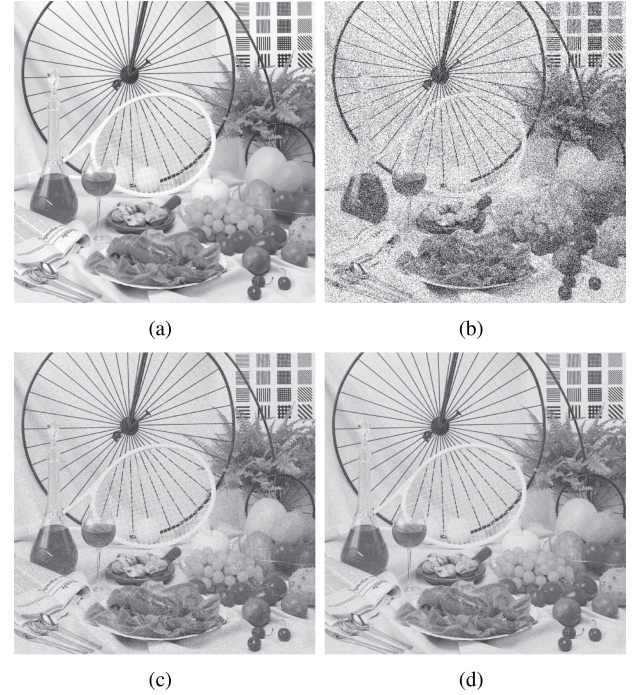


Fig. 6. Results of image fusion on a 512×512 grayscale picture using the PTCR-ZNN model (10). (a) Original image. (b) Noisy image with SNR = 1 dB. (c) Fused image when $S = 20$. (d) Fused image when $S = 50$.

PTCR-ZNN model (10) is then used to acquire its optimal solution $\mathbf{x}^*$. Eventually, the fused image $\mathbf{P}^*(i, j)$ is reconstructed via $\mathbf{P}^*(i, j) = \mathbf{z}^*((i-1)J + j)$.

To perform image fusion using the PTCR-ZNN model (10), we select a grayscale picture with a size of 512×512 , which is shown in Fig. 6(a) and has the highest quality. Then, a certain number of independent noisy pictures with Gaussian noises are simulatively generated; their sizes are 512×512 , while their signal-to-noise ratios (SNRs) are 1 dB. One example of noisy images can be found in Fig. 6(b). Moreover, we have tested fusion on different number of noisy images, and the final optimal fused images obtained by the PTCR-ZNN model (10) are demonstrated in Fig. 6(c) and (d). Evidently, this application has notably improved the quality of input noised images, and the denoising effect becomes better with larger S .

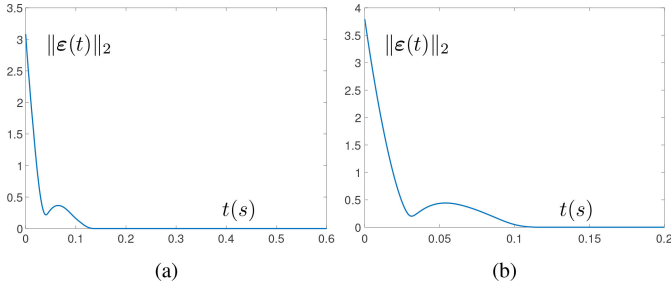


Fig. 7. Computation error $\|\varepsilon(t)\|_2$ synthesized by the PTCR-ZNN model (9) during image fusion with different S . (a) $S = 20$. (b) $S = 50$.

We set $\gamma_1 = \gamma_2 = 5$, $r_1 = 2$, and $r_2 = 0.5$ in the PTCR-ZNN model (9) for showing that the predefined-time convergence of such a neural network can be accelerated by adopting larger γ_1 and γ_2 . As can be seen in Fig. 7, when $S = 20$ or $S = 50$, $\|\varepsilon(t)\|_2$ drops to 0 before the time upper bound 0.6 s that we calculated according to Theorem 3. Besides, the matrix $\mathbf{G}$ in (32) is a high-dimensional matrix when $S = 20$ and $S = 50$. However, our novel PTCR-ZNN model (9) has successfully solved such a high-dimensional QP problem. The image fusion denoising method can be used in more practical industrial applications. For this reason, we have tested our PTCR-ZNN-model-based image fusion method in magnetic resonance (MR) image denoising tasks. In this experiment, we use high-quality MR images of human brain downloaded from online database BrainWeb¹. The downloaded MR image datasets are T1-MR with slice sickness being 1 mm, and they have 181 slices with a size of 217×181 . Then, two different types of noises are applied to original MR images: one is the aforementioned zero-mean Gaussian noise, and the other is the Rician noise (because it appears in MR images more frequently). The Rician noise is synthesized from Gaussian noise, but it happens in the complex domain [27]. For other model parameters, we set the same as $\gamma_1 = \gamma_2 = 5$, $r_1 = 2$, $r_2 = 0.5$, and $S = 100$. We can see in Fig. 8 that when the applied noise level is 20%, both the Gaussian-noise-polluted MR images or Rician-noise-polluted MR images have bad image quality and are unrecognizable. However, Fig. 8(c) and (f) demonstrates that our PTCR-ZNN-model-based image fusion has improved qualities of noisy MR images a lot, and the fused images are recognizable. In addition, the corresponding errors $\|\varepsilon(t)\|_2$ are plotted in Fig. 9, whose decreasing trajectories match our expectation perfectly.

VIII. APPLICATION TO ROBOT CONTROL

Like the image fusion shown in Section VII, there are still many applications that can take advantage of our PTCR-ZNN model (10), including controlling redundant robotic manipulators. Generally speaking, extra degrees of freedom make redundant robot manipulators capable of avoiding obstacles while accomplishing tasks of the end-effector. However, such characteristic makes the inverse kinematics control of a redundant robotic manipulator a difficult job, especially when considering joint limits of the manipulator. In this section, we will use

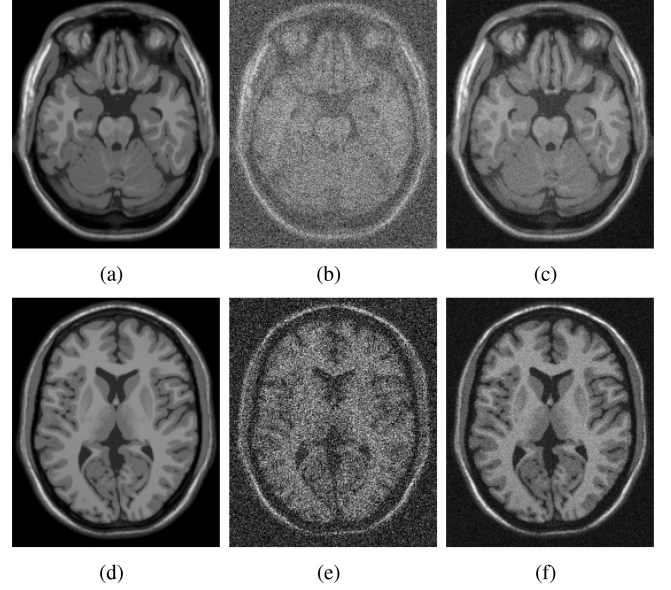


Fig. 8. Results of image fusion on T1-MR images using the PTCR-ZNN model (10). (a) Slice 50 of T1-MR. (b) Slice 50 polluted by Gaussian noise. (c) Fused image of noise-polluted slice 50 when $S = 100$. (d) Slice 80 of T1-MR. (e) Slice 80 polluted by Rician noise. (f) Fused image of noise-polluted slice 80 when $S = 100$.

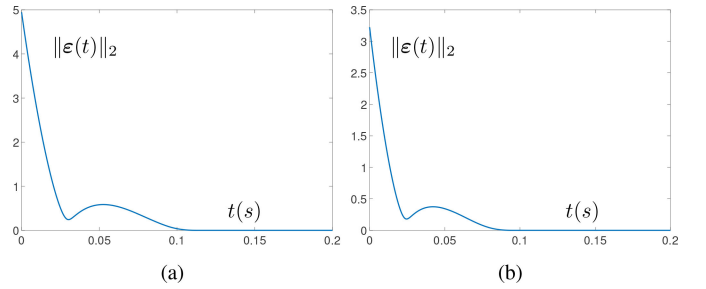


Fig. 9. Residual error $\|\varepsilon(t)\|_2$ synthesized by the PTCR-ZNN model (10) during image fusion when using different T1-MR images. (a) Using Gaussian-noise-polluted Slice 50. (b) Using Rician-noise-polluted Slice 80.

our PTCR-ZNN model (10) to control a redundant robot arm PUM560 in the simulation experiments.

Inspired by [18], we adopt the repetitive motion planning method that was solved at velocity level with joint limits considered. This scheme is based on the TVQPEI (27)–(29) problem and is expressed as follows:

$$\begin{aligned} & \min \dot{\boldsymbol{\theta}}(t)^T \dot{\boldsymbol{\theta}}(t) / 2 + \mathbf{q}^T(t) \dot{\boldsymbol{\theta}}(t) \\ & \text{s.t. } \mathbf{W}_J(\boldsymbol{\theta}(t)) \dot{\boldsymbol{\theta}}(t) = \dot{\mathbf{r}}_d(t) + \sigma(\mathbf{r}_d(t) - \mathcal{F}(\boldsymbol{\theta}(t))) \\ & \quad \boldsymbol{\theta}^-(t) \leq \boldsymbol{\theta}(t) \leq \boldsymbol{\theta}^+(t) \\ & \quad \dot{\boldsymbol{\theta}}^-(t) \leq \dot{\boldsymbol{\theta}}(t) \leq \dot{\boldsymbol{\theta}}^+(t) \end{aligned} \quad (34)$$

where $\boldsymbol{\theta}(t) \in \mathbb{R}^6$, $\dot{\boldsymbol{\theta}}(t) \in \mathbb{R}^6$ are the joint angle vector and the joint velocity vector of PUM560, respectively, and $\mathbf{q}(t) = \xi(\boldsymbol{\theta}(t) - \boldsymbol{\theta}(0))$ denotes the joint drift between current and initial states with $\xi > 0$. Besides, $\mathbf{r}_d(t) \in \mathbb{R}^3$ is given

¹[Online]. Available: <https://brainweb.bic.mni.mcgill.ca/brainweb/>

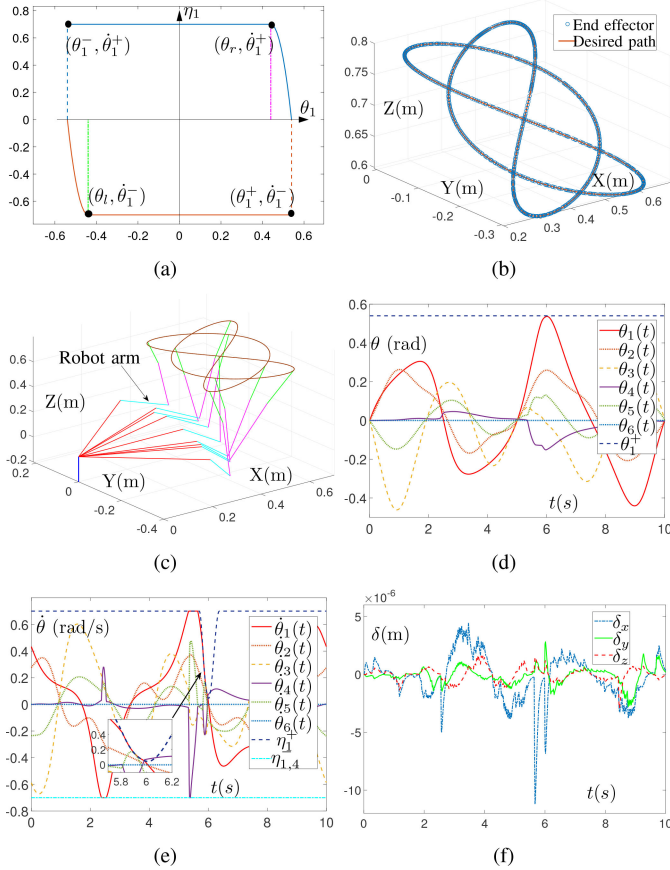


Fig. 10. Simulation results of using the PTCR-ZNN model (10) to control PUM560 robot manipulator to track the Lissajous curve path. (a) Relationship between θ_1 and η_1 . (b) Trajectory of the end-effector and the desired path. (c) Full view of path tracking with robot arm in Cartesian space. (d) Traces of joint angles. (e) Traces of joint velocities. (f) Position errors of the end-effector in three dimensions.

nonstationary desired path vector of the end-effector in 3-D space, $\mathcal{F}(\cdot) \in \mathbb{R}^6 \rightarrow \mathbb{R}^3$ is the end-effector position calculation function, and $\mathbf{W}_J(t) = \mathbf{W}_J(\boldsymbol{\theta}(t)) = \partial \mathcal{F}(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$ represents the Jacobian matrix. Then, $\sigma(\mathbf{r}_d(t) - \mathcal{F}(\boldsymbol{\theta}(t)))$ with $\sigma > 0$ is used for feedback control. In addition, $\boldsymbol{\theta}^+(t), \boldsymbol{\theta}^-(t) \in \mathbb{R}^6$ and $\dot{\boldsymbol{\theta}}^+(t), \dot{\boldsymbol{\theta}}^-(t) \in \mathbb{R}^6$ denote upper and lower bounds of joint angle and joint velocity, respectively. In our TVPQEI problem (1)–(3), only one unknown vector $\mathbf{x}(t)$ is allowed, so limitation of $\boldsymbol{\theta}(t)$ is integrated into $\dot{\boldsymbol{\theta}}(t)$ limit. Finally, the control scheme (34) is rewritten into the standard TVPQEI problem (1)–(3) with input coefficients defined as $\mathbf{x}(t) = \dot{\boldsymbol{\theta}}(t)$, $\mathbf{S}(t) = \mathbf{I}_6$, $\mathbf{p}(t) = \mathbf{q}(t)$, $\mathbf{A}(t) = \mathbf{W}_J(t)$, $\mathbf{b}(t) = \dot{\mathbf{r}}_d(t) + \sigma(\mathbf{r}_d(t) - \mathcal{F}(t))$, and $\mathbf{C}(t) = [\mathbf{I}_6, -\mathbf{I}_6]^T$, where $\mathbf{I}_n$ stands for the $n \times n$ identity matrix size. We define the last coefficient as $\mathbf{d}(t) = [\boldsymbol{\eta}^{+T}, \boldsymbol{\eta}^{-T}]^T$ with

$$\eta_i^+ = \begin{cases} \dot{\theta}_i^+, & \theta_i(t) \in [\theta_i^-, \theta_r] \\ \phi_1(\theta_i(t)), & \theta_i(t) \in [\theta_r, \theta_i^+] \end{cases}$$

$$\eta_i^- = \begin{cases} \dot{\theta}_i^-, & \theta_i(t) \in [\theta_l, \theta_i^+] \\ \phi_2(\theta_i(t)), & \theta_i(t) \in [\theta_i^-, \theta_l] \end{cases}$$

where $\phi_1(\theta_i) = -\dot{\theta}_i^+(\theta_i - \theta_r)^2 / (\theta_i^+ - \theta_r)^2 + \dot{\theta}_i^+$ and $\phi_2(\theta_i) = -\dot{\theta}_i^-(\theta_i - \theta_l)^2 / (\theta_i^- - \theta_l)^2 + \dot{\theta}_i^-$. In the above equations, $\theta_i, \dot{\theta}_i$, and η_i are the i th ($\forall i \in 1, 2, \dots, 6$) element of vectors $\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}$, and $\boldsymbol{\eta}$, respectively, and $\theta_r, \theta_l \in [\theta_i^-, \theta_i^+]$ satisfying $\theta_r > \theta_l$ are design parameters. Therefore, relationships between θ_i and $\dot{\theta}_i$ have been shown in Fig. 10(a).

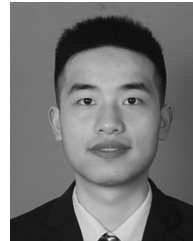
We now use the above PTCR-ZNN model (10) based inverse kinematics control method to control PUM560 to track a rotated Lissajous curve path in 3-D space. Parameters of the PTCR-ZNN model (10) are selected as $\gamma_1 = \gamma_2 = 1, r_1 = 2, r_2 = 0.8$, and $\delta^+ = 10^8$. Specifically, joint limits in $\mathbf{d}(t)$ are set to be $\boldsymbol{\theta}^+ = [0.54, 1, 1, 1, 1, 1]$ rad, $\dot{\boldsymbol{\theta}}^+ = 0.7 * [1, 1, 1, 1, 1, 1]$ rad/s, $\theta_r = \theta_i^+ - 0.1$ rad, and $\boldsymbol{\theta}^- = -\boldsymbol{\theta}^+, \dot{\boldsymbol{\theta}}^- = -\dot{\boldsymbol{\theta}}^+, \theta_l = \theta_r$. The results of path tracking in 10 s are depicted in Fig. 10. From Fig. 10(a), taking θ_1 and η_1 for example, it is clear that two functions $\phi_1(\cdot)$ and $\phi_2(\cdot)$ have smoothed boundaries of $\dot{\theta}(t)$, making $\mathbf{d}(t)$ differentiable, which is required by our PTCR-ZNN model. Then, let us see Fig. 10(d); there is a time period when the joint angle θ_1 is very close to the limit $\theta_1^+ = 0.54$ rad and drops back at about $t = 6$ s. Correspondingly, we find in Fig. 10(e) that the actual joint velocity limit η_1^+ begins decreasing when θ_1 approaches θ_r and decreases nearly to zero at about $t = 6$ s, preventing θ_1 from crossing the bound. Apart from $\theta_1, \dot{\theta}_4$ also hit the velocity limit η_4^- at about $t = 5.5$ s. However, in Fig. 10(d) and (e), limits of joint angle and joint velocity have all been satisfied well. Then, the end-effector trajectory and the desired path are illustrated in Fig. 10(b), while the positions of all arms are shown in Fig. 10(c), where the actual path and the desired path overlaps very well. As in Fig. 10(f), the end-effector position errors of three coordinates are at level of 10^{-6} m. Thus, we conclude that the PUM560 tracks the Lissajous curve path with high precision even with strict limits. The superiorities of our PTCR-ZNN model (10) have now been proved by above robot control simulation results.

IX. CONCLUSION

In this article, by designing a special PPAF and combining it with the noise-tolerant enhanced ZNN model, we proposed a PTCR-ZNN model that solves different branches of QP problems in a unified model. Theoretical analysis on the predefined-time convergence and the noise suppression ability against different time-varying noises were discussed and proved. Results of simulation experiments validated that the PTCR-ZNN model can obtain accurate solutions of TVQP problems in designated finite time and maintain high accuracy when facing with large perturbations. In summary, our proposed PTCR-ZNN model (10) was broadly suitable for various QP problems with tunable and elastic convergence time, and it was also reliable in complex noisy environments. Besides, the PTCR-ZNN model can handle real-world applications such as image fusion and robotic manipulators control. The future investigation direction is to further enhance the convergence or robustness performance of the PTCR-ZNN model (9) and to apply it in more realistic problems such as thermal imaging CFRP defect detection in [28].

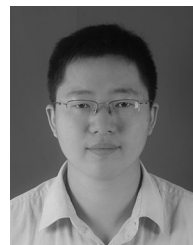
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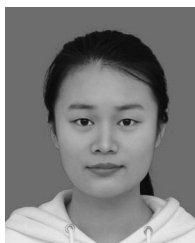
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